Lecture 4 Live Problems

Vector fields and flow. Trajectories in state space.

Points toward lab manual pages: **ST6, ST7, ST8** (section L1 S3)

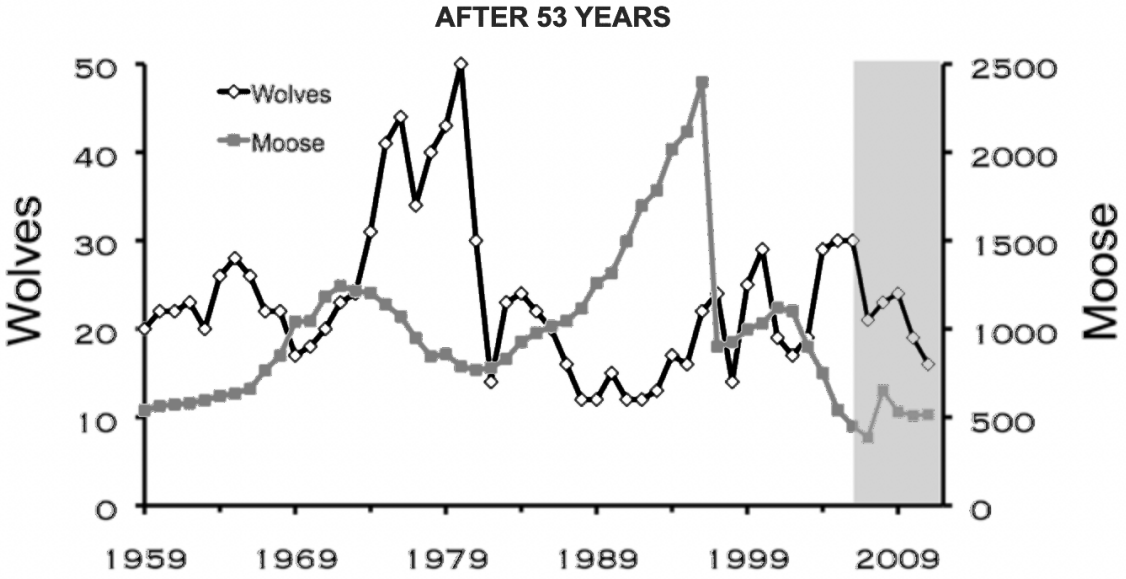
1. Reading the plot. Wolf and moose populations on Isle Royale, 1959–2012. Notice that the two curves use different vertical axes.

(a) Which axis do you use to read the number of wolves? The number of moose?

(b) Approximately how many moose were there in 1979? _____

(c) Approximately how many wolves were there in 1979? _____

(d) Write your answers as an ordered pair (M, W). _____



Data from isleroyalewolf.org.

2. Moose–wolf space. Let M be the number of moose and W the number of wolves on Isle Royale.

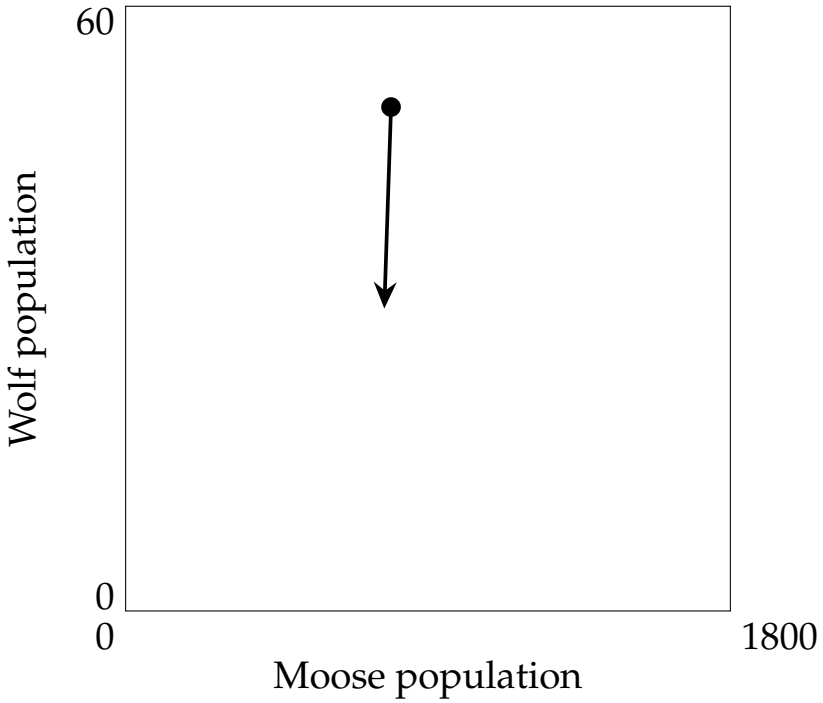
- 1980: 790 moose and 50 wolves.
- 1981: 770 moose and 30 wolves.
- 1982: 780 moose and 14 wolves.

From 1981 to 1982:

(a) What is ΔM? _____ What is ΔW? _____

(b) What is the change vector? _____

(c) The change from 1980 to 1981 is drawn below. Draw the change vector from 1981 to 1982.



3. Modeling predators and prey. Our basic assumptions about moose and wolves on Isle Royale:

- More moose = more wolf food, good for wolves!
- More wolves = more moose eaten, bad for moose!
- Moose would happily flourish, eating grass and stuff.
- Wolves need to eat some big prey. Otherwise they will die out.

Make a diagram to illustrate the influences of the quantities M and W on themselves and each other.